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Please, answer the theoretical questions in detail, using only the provided sheets. The exam is **open-book**, and you can use **no electronic devices** except for a portable calculator. You are encouraged to explain your reasoning, like in an oral exam. You are also encouraged to draw sketches for concepts that are visual in nature. Please refrain from copying from the notes! Use your own words and drawings.

The exam duration is **2hr**.

This written exam gives up to 32 points. The lab assignments provide up to extra 4 points. Total grades beyond 30/36 will be graded with 30 with laude.

Exercise 1

- A) Plot the derivative of the function $f(x) = x^2/3$ if $x > 0$; $f(x) = 0$ if $x \leq 0$.
- B) Explain the difference between a linear and a non-linear classifier. Which one is more powerful in solving classification problems? Justify your answer.
- C) Explain the concept of “learning rate” as used in the SGD algorithm
- D) Consider a fully connected neural network with 2 layers: 4 inputs, 16 internal neurons and 3 output neurons. How many weights does it have?

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Exercise 2

Consider a visual transformer (ViT) where a 64x64 grayscale image is split in 8x8 non-overlapping patches. Each patch is passed through a convolutional layer with 3x3 filters (zero padding is used) and 3 output channels with RELU non linearity. The outputs channels 1,2,3 of the convolutional layer are unrolled and become the rows (dimension d) of the Q (output channel 1), K (output channel 2), V (output channel 3) matrices of the attention layer. The number of rows (seq) is equal to the number of patches.

- A) Compute d, seq and the number of elements of QK^T
- B) Compute the total number of MAC in the convolutional layer and in the attention layer
- C) Compute the total storage needed to store the convolutional weights as well as the Q, K, V matrices assuming that activations are stored as FP32 and weights are stored as FP16.
- D) Compute the total number of exp function evaluations and RELU evaluation needed for the ViT.

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Exercise 3

A parallel cluster of 4 RISC-V processors is executing a convolutional layer with 4 input channels (32x32) and 18 output channels with 3x3 filters, followed by average pooling (2x2) and by RELU nonlinearity. Zero padding is used when computing the convolutions. The computation is parallelized assigning output channels to processors. Activations are stored in 8 bits and weights are stored in 4 bits. Clock is at 100MHz.

III.a) Compute the L1 minimal shared memory size that is needed to store the input and output channels, and the weights.

III.b) Assume that the IPC (instructions per clock) is 0.9 when computing convolutions and 0.8 when computing averaging and RELU. Consider the number of instructions equal to 1.2 the number of MACs, additions and divisions, RELU computations for the convolutional, average pooling and RELU computation steps, respectively. Compute the total sequential execution time using a single processor

III.c) Compute parallel execution time. Assume that there are global barriers after convolutional, average pooling and RELU steps.

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